

APPROVED METHOD:
DIRECTIONAL POSITIONING OF
PHOTOMETRIC DATA
AN AMERICAN NATIONAL STANDARD

ANSI/IES LM-72-20

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has been approved by the IES.
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be directed to the IES.

**Prepared by the
IES Testing Procedures Committee**



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Approved by the IES Standards Committee February 13, 2020 as a Transaction of the Illuminating Engineering Society.

Approved April 21, 2020 as an American National Standard.

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Published by the Illuminating Engineering Society, 120 Wall Street, New York, New York 10005.

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Printed in the United States of America.

ISBN# 978-0-87995-215-0

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1.0 Introduction and Scope

1.1 Introduction

This document is intended to provide a clearly defined set of terms to be used to designate the angular positioning of photometric data. The terms are intended to be used by both providers and users of lighting software so as to unambiguously and consistently specify how photometric data is to be "rotated" before it is used.

Two similar but fundamentally irreconcilable types of positioning have existed for years. The "Clockwise from North" ("CN") system starts by rotating luminaire photometric data clockwise from north. The "Counter-Clockwise from East" ("CCE") system starts by rotating luminaire photometric data counter-clockwise from east. A third positioning method, called "aiming luminaire at a point," has also existed for years and will also be discussed.

Many early and current lighting programs use north as the starting point for luminaire orientation, with angles measure clockwise, as on a compass. Specifiers who expected pole-top installers to use compasses would often provide magnetic bearings as well. Users (designers, engineers, and steeplejacks) accustomed to these techniques would choose the CN system.

Many computer-aided design (CAD) programs and many related lighting programs use east as the starting point for orientation, with angles measured counterclockwise. Users (designers, engineers, draftsmen, physicists, and mathematicians) accustomed to these methods would select the CCE system.

Designers of floodlighting, security, and sports lighting often use the notion of "aiming a luminaire at a point," so most computer programs that handle floodlighting and sports lighting employ the "aiming at a point" concept. For floodlighting and sports lighting work, aiming at a point is often the most convenient approach to choose.

1.2 Scope

It is important to note that this document assigns

specific, standard meaning to certain words; thus, some users might need to adjust their vocabulary. Most should find a positioning system compatible with their normal patterns and experience. This document does not show a preference for any system, but rather describes each of the three systems and then explains how to convert between them.

In applying CN or CCE, four types of angular "rotation" are defined (Orient, Tilt, Roll, and Spin), and each shall be preformed in the defined order. Any or all of the rotation angles may be zero. Each subsequent rotation is applied to the axis system as previously rotated (the local axes), not the original axis system (the global axes).

Orientation, Tilt, Roll, and Spin can be described using X, Y, and Z axes, but the CN system and CCE system are described separately, and separate drawings of the coordinate system are needed.

2.0 Photometry Systems

Most photometry is available in a standard format in North America.¹ Two types of photometry, which differ in the orientation of the photometric web, have traditionally been used with that standard. Type C is commonly used for indoor and roadway luminaires.^{2,3} Type B photometry has been used for adjustable floodlights.⁴ (Refer to **Annex B** for a description of Type B and Type C photometry.)

(Note: The figures discussed in Section 3.0 will assist in understanding the positioning systems described in **Sections 2.1** and **2.2.**)

2.1 Clockwise from North (CN) System

The CN system angular positioning system is applicable to both Type B photometry and Type C photometry. With this system, the initial position of photometric zero (horizontal angle = 0°, vertical angle = 0°) is straight down in the direction of the -Z axis. To set the initial position of the luminaire photometric grid in three-dimensional space, a second photometric

direction is established. The photometric direction (0° , 90°), where the horizontal angle = 0° and the vertical angle = 90° , is convenient for both Type B and Type C photometry. The initial photometric direction (0° , 90°) for the CN system is placed in the direction of the +Y axis (north).

First, *Orientation* is applied about the Z axis clockwise, looking in the direction of the -Z axis. Positive angles move the +Y axis toward the +X axis. Orientation = 0° places the +Y axis toward north. (Note: Initially, before any rotation is applied, the local and global coordinate systems are aligned.)

Second, *Tilt* is applied about the local X axis clockwise, looking in the direction of the local +X axis. Positive angles move the local -Z axis toward the local +Y axis. (Moving the local +Y axis toward the local +Z axis is equivalent.)

Third, *Roll* is applied about the local y axis clockwise, looking in the direction of the local +Y axis. Positive angles move the local -Z axis toward the local -X axis. (Moving the local +Z axis toward the local +X axis is equivalent.)

Fourth, *Spin* is applied about the local Z axis clockwise, looking in the direction of the local -Z axis of the luminaire photometric data. Positive angles move the local +Y axis toward the local +X axis.

2.2 Counterclockwise from East (CCE) System

The CCE system angular positioning system is applicable to both Type B photometry and Type C photometry. With this system, the initial position of photometric zero (horizontal angle = 0° , vertical angle = 0°) is straight down in the direction of the -Z axis. To set the initial position of the luminaire photometric grid in three-dimensional space, a second photometric direction is established. The photometric direction (0° , 90°), where the horizontal angle = 0° and the vertical angle = 90° , is convenient for both Type B and Type C photometry. The initial photometric direction (0° , 90°) for the CCE system is placed in the direction of the +X axis (east).

First, *Orientation* is applied about the Z axis counterclockwise, looking in the direction of the -Z axis.

Positive angles move the +X axis toward the +Y axis. Orientation = 0° places the +X axis toward east. (Note: Initially, before any rotation is applied, the local and global coordinate systems are aligned.)

Second, *Tilt* is applied about the local Y axis clockwise, looking in the direction of the local -Y axis. Positive angles move the local -Z axis toward the +X axis (Moving the +X axis toward the local +Z axis is equivalent).

Third, *Roll* is applied about the local X axis clockwise, looking in the direction of the local +X axis. Positive angles move the local -Z axis toward the local +Y axis.

Fourth, *Spin* is applied about the local Z axis counterclockwise, looking in the direction of the local -Z axis. Positive angles move the local +X axis toward the local +Y axis.

3.0 CN and CCE Positioning System Illustrations

Figure 3-1 shows, for both systems, each individual rotation. Each diagram shows local axes and a luminaire (rectangular block) *before* the rotation. The arrow indicates the direction of positive rotation. The arrow length represents 90° .

Figure 3-2 shows an example of angular positioning for each system. For the CN system, the angles are: Orientation = 90° , Tilt = 45° , Roll = 90° , and Spin = 45° . For the CCE system, the angles are: Orientation = 0° , Tilt = 45° , Roll = 90° , and Spin = -45° . Each diagram shows the local axes and a "luminaire" after the rotation is applied. Each diagram shows the local axes and a luminaire (rectangular block) *after* the rotation has been applied.

It is important to note that other sets of four angles achieve the same position. For the CN system, Orientation = 90° , Tilt = 0° , Roll = 90° , and Spin = 0° suffice. For the CCE system, Orientation = 0° , Tilt = 0° , Roll = 90° , and Spin = 0° suffice.

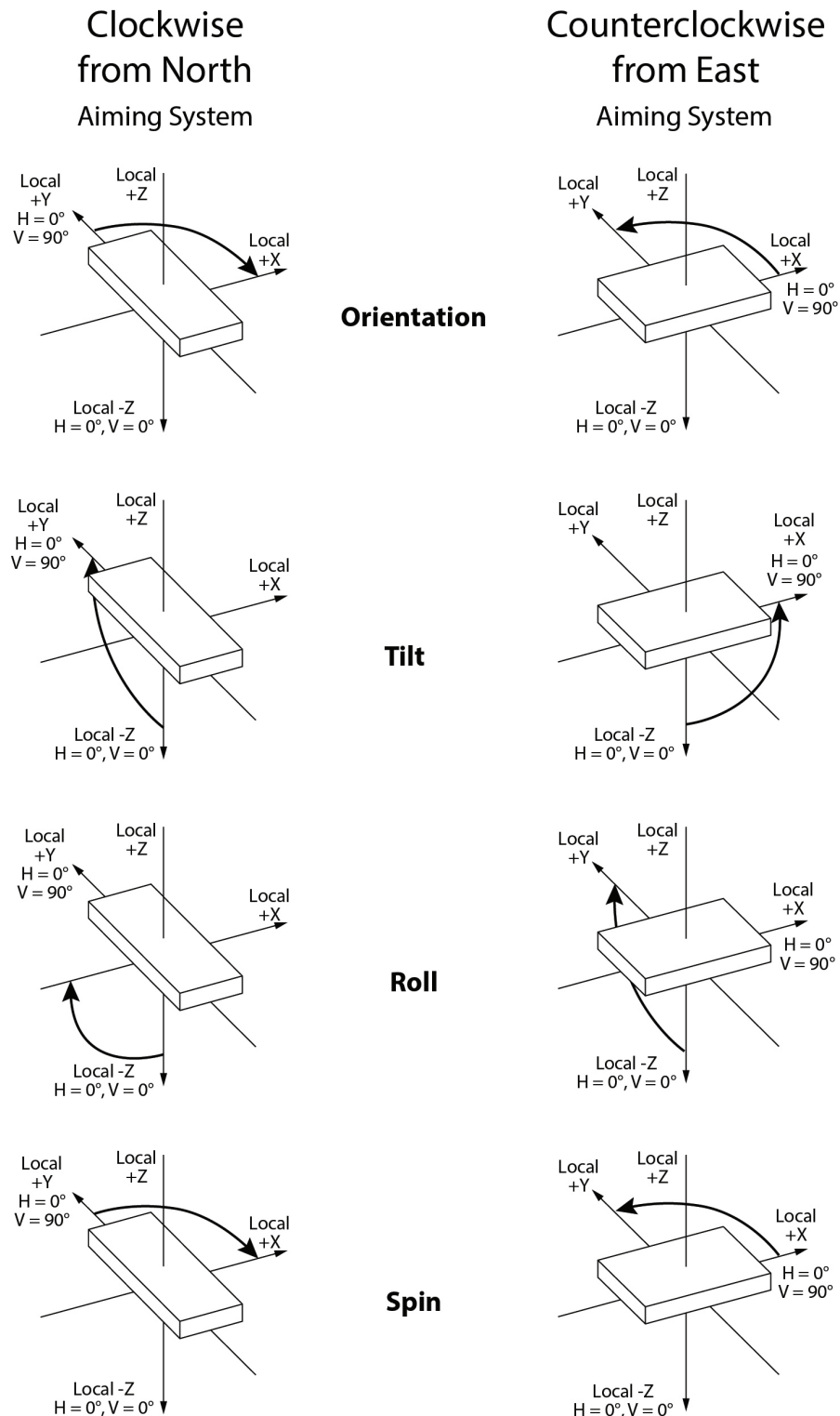


Figure 3-1. Positioning definitions for a luminaire (rectangular block) before rotation in the CN aiming system (left column) and the CCE aiming system (right column). Each curved arrow indicates the direction of positive rotation, with the arrow length equal to 90°. (© Illuminating Engineering Society)

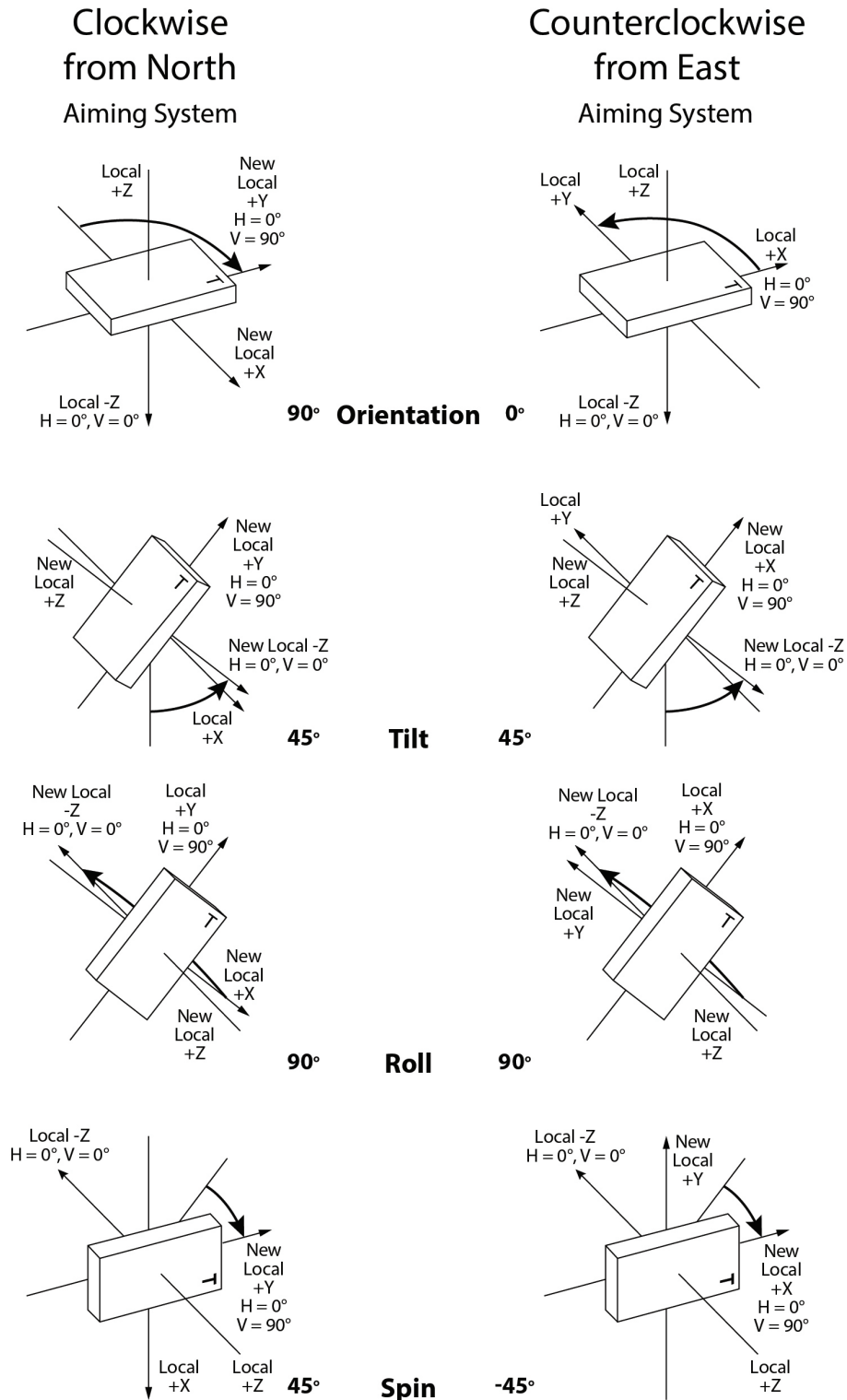


Figure 3-2. Equivalent positioning examples for a luminaire (rectangular block) after rotation in the CN aiming system (left column) and the CCE aiming system (right column). One corner of the luminaire is marked with the symbol "T" to help the reader visualize the positioning process. (© Illuminating Engineering Society)

4.0 Comments for Program Developers

All possible positions can be achieved by omitting *Roll* or by omitting *Spin*. Historically, programs have been set up to use *Orientation/Tilt/Spin* or *Orientation/Tilt/Roll* (though often using different terminology). Each of these systems is a subset of the *Orientation/Tilt/Roll/Spin* system defined in this document.

All possible positions can be achieved in the CN system and in the CCE system.

The physical structure of some mounting mechanisms might make achieving certain positions impossible.

Program developers may choose to implement only one system and may choose to omit either *Roll* or *Spin*, as noted above. The terms *Orientation*, *Tilt*, *Roll*, and *Spin* shall be used to specify luminaire photometric positioning only as defined here, in the order defined. The use of previous terms such as "rotation" and "bearing" for *Orientation* and "rotation" and "cant" for *Roll* is deprecated.

Program developers are encouraged to let the user choose between using the CN system and the CCE system. Program developers are also encouraged to let the user enter all four rotation angles or to let the user choose which of the four rotation angles are to be entered. The order in which angles are applied is fixed and cannot be configured.

This standard is intended to be convenient for specifiers and other users. Almost all users can find a positioning method compatible with their accustomed techniques. The advantage of adhering to a positioning standard is that ambiguous in rotation direction is removed, and specification of aiming is more reliable.

Caution to users of any positioning system: Positioning Type B and asymmetrical Type C photometry requires careful checking of the photometry and/or the results, since manufacturers' and laboratories' interpretations of the standards have varied. Luminaires such as wall-washers, sconces, step lights, and in-wall uplights may

produce unexpected results, depending on the test position. For example, fluorescent wall-wash luminaires may have the 0° plane parallel to the lamps, or the 0° plane may be perpendicular to the lamps to represent the line of symmetry. The 0° plane may (or may not) contain higher candela values than the 180° plane. Further, some wall-wash luminaires have higher candela values away from the wall rather than toward it. As another example, uplights may have been tested with photometric zero being straight up rather than straight down.

5.0 Aiming a Luminaire at a Point

The "aiming a luminaire at a point" positioning system is defined in this document as directing the nadir (local -z axis) from the luminaire photometric center to the aiming point, keeping the 0° plane vertical. Assuming this definition is met, *Orientation* and *Tilt* (as previously defined) are the only angles needed to duplicate a position achieved by "aiming a luminaire at a point." Other definitions of "aiming a luminaire at a point" are deprecated.

It is important to note that just aiming a luminaire at a point (or using only *Orientation* and *Tilt* angles) might not allow all possible photometric orientations. With the added angular parameter *Spin* (as defined), all possible photometric positions can be achieved. *Spin* is applied about the local Z axis, either clockwise (as done in the CN system), or counterclockwise (as is done in the CCE system) looking in the direction of the local -Z axis of the luminaire photometric data.

For computer programs allowing both angular positioning and "aiming a luminaire at a point — with *Spin*," the direction of *Spin* shall be consistent. (That is, if CN is required by the program or chosen by the user, *Spin* shall be clockwise. If CCE is required by the program or chosen by the user, *Spin* shall be counterclockwise.) When only the "aiming a luminaire at a point — with *Spin*" method is used, *Spin clockwise* or *Spin counterclockwise* shall be indicated.

For the “aiming a luminaire at a point – with Spin” positioning system, using terms other than *Spin* for a final rotation about the local Z axis is deprecated.

When the aiming point is directly above or below the luminaire photometric center, the luminaire position should be equivalent to Orientation = 0°, Tilt = 0°, (for below), or Tilt = 180° (for above).

Figure 5-1 illustrates how “aiming a luminaire at a point – with Spin” can be equivalent to either CN or CCE aiming. A luminaire is shown aimed along a vector (1, 0, -1) from photometric center with no (0°) spin. Any point along that vector from photometric center is equivalent. For the CN system, the equivalent angles are Orientation = 90°, Tilt = 45°. For the CCE system, the equivalent angles are Orientation = 0°, Tilt = 45°. It should be noted that the luminaire position in **Figure 5-1** matches the luminaire position in **Figure 3-2**

(see **Section 3**) after Orientation and Tilt have been applied.

Figure 5-2 illustrates how “aiming a luminaire at a point – with Spin” can be equivalent to either CN or CCE aiming. A luminaire is shown aimed along a vector (0, 1, 0) from the photometric center, with Spin = 90° for compatibility with the CN system and Spin = -90° for compatibility with the CCE system. For the CN system, equivalent angles are Orientation = 0°, Tilt = 90°, Spin = 90°. For the CCE system, equivalent angles are Orientation = 90°, Tilt = 90°, Spin = -90°. It should be noted that the final position of the luminaire in **Figure 5-2** matches the final position of the luminaire in **Figure 3-2** (see **Section 3**).

It should be noted that *Roll* has not been used in the description of “aiming a luminaire at a point.” Roll angles would always be 0° in equivalent aiming positions.

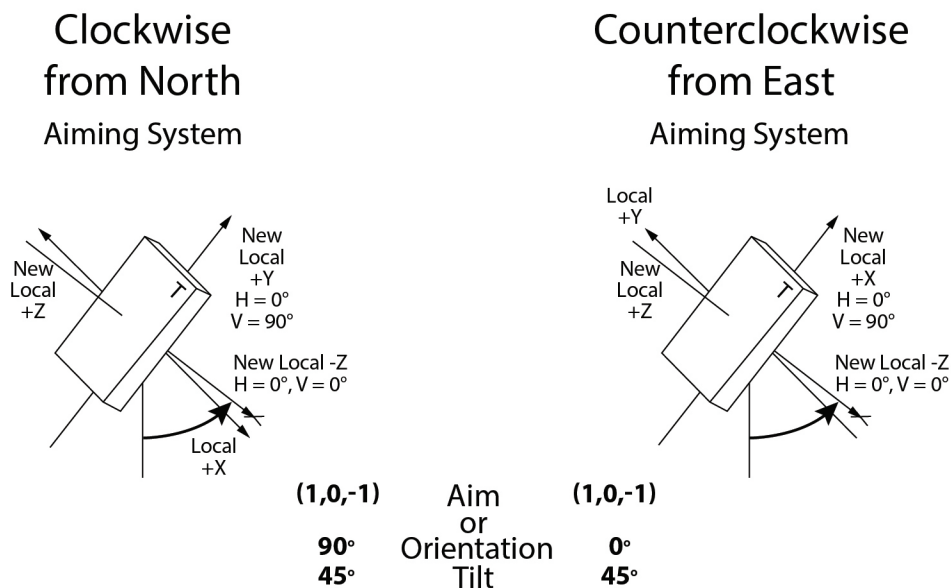


Figure 5-1. Example of “aiming a luminaire at a point – without Spin” where the luminaire (rectangular block) is aimed along a vector (1, 0, -1) from photometric center with Spin = 0°. Positioning equivalence with the CN system (left) or the CCE system (right) is obtained when the indicated Orientation and Tilt angles are present for these two systems. One corner of the luminaire is marked with the symbol “T” to help the reader visualize the positioning process. (© Illuminating Engineering Society)

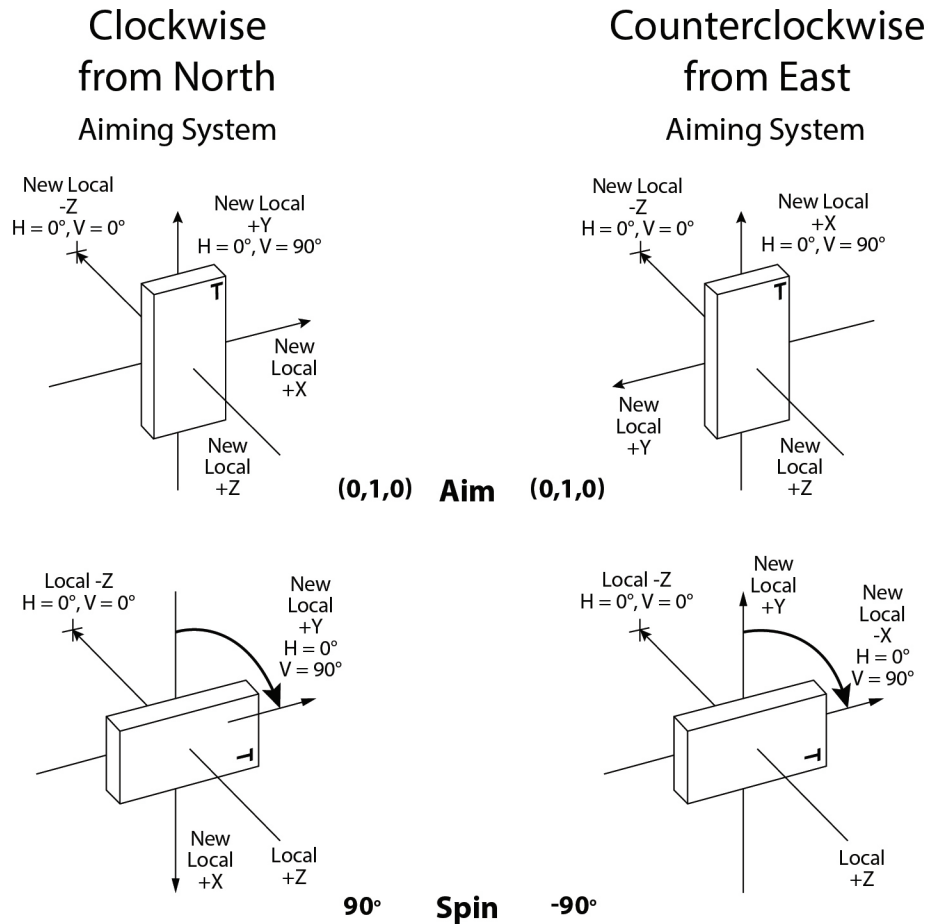


Figure 5-2. Example of “aiming a luminaire at a point – with Spin” where the luminaire (rectangular block) is aimed along a vector (0, 1, 0) from photometric center. When Spin = 90°, (clockwise), compatibility is maintained with the CN system (left column). When Spin = -90°, (counterclockwise), compatibility is maintained with the CCE system (right column). One corner of the luminaire is marked with the symbol “T” to help the reader visualize the positioning process. (© Illuminating Engineering Society)

Annex A – Formulae for Aiming Systems

This Annex contains formulae that implement both the CN and CCE aiming standards. Other approaches, coordinate systems, and formulae (such as transformation matrices⁵⁻⁷) can also be used. By simplifying the implementation of this standard, it is intended that developers will find this Standard easy to adopt.

A.1 Symbols

O is the orientation angle

T is the tilt angle

R is the roll angle

S is the spin angle

H is the horizontal photometric angle

V is the vertical photometric angle

X_c, Y_c, Z_c are the "global" coordinates in the space with origin at the luminaire (photometric) center.

X_p, Y_p, Z_p are local coordinates relative to the luminaire's "photometric coordinate system" (which match X_c, Y_c, Z_c when the aiming angles are all 0). The Origin is at photometric center.

A.2 Clockwise from North (CN) Aiming

When using Type C photometry with CN aiming, the following equations will convert between photometric angles and the photometric coordinate system:

$$H = \text{atan2}(-X_p, Y_p) \text{ (See Section A.7, Notes 2, 3, 5.)}$$

$$V = \text{atan2}[\sqrt{X_p^2 + Y_p^2}, -Z_p] \text{ (See Section A.7, Note 3.)}$$

$$X_p = -\sin H \sin V$$

$$Y_p = \cos H \sin V$$

$$Z_p = -\cos V$$

When using Type B photometry with CN aiming, the following equations will convert between photometric angles and the photometric coordinate system:

$$H = \text{atan2}[X_p, \sqrt{Y_p^2 + Z_p^2}] \text{ (See Section A.7, Notes 2, 3)}$$

$$V = \text{atan2}(Y_p, -Z_p) \text{ (See Section A.7, Notes 3, 4)}$$

$$X_p = \sin H$$

$$Y_p = \cos H \sin V$$

$$Z_p = -\cos H \cos V$$

Conversions from space coordinates to photometric coordinates with CN aiming can be made using the following equations:

$$\begin{aligned} X_p = & X_c[(\cos O \cos R + \sin O \sin T \sin R)\cos S - \sin O \cos T \sin S] + \\ & Y_c[(-\sin O \cos R + \cos O \sin T \sin R)\cos S - \cos O \cos T \sin S] + \\ & Z_c(-\cos T \sin R \cos S - \sin T \sin S) \end{aligned}$$

$$\begin{aligned} Y_p = & X_c[(\cos O \cos R + \sin O \sin T \sin R)\sin S + \sin O \cos T \cos S] + \\ & Y_c[(-\sin O \cos R + \cos O \sin T \sin R)\sin S + \cos O \cos T \cos S] + \\ & Z_c(-\cos T \sin R \sin S + \sin T \cos S) \end{aligned}$$

$$\begin{aligned} Z_p = & X_c(\cos O \sin R - \sin O \sin T \cos R) + \\ & Y_c(-\sin O \sin R - \cos O \sin T \cos R) + \\ & Z_c(\cos T \cos R) \end{aligned}$$

Conversions from photometric coordinates to space coordinates with CN aiming can be made using the following equations:

$$\begin{aligned} X_c = & X_p[(\cos O \cos R + \sin O \sin T \sin R)\cos S - \sin O \cos T \sin S] + \\ & Y_p[(\cos O \cos R + \sin O \sin T \sin R)\sin S + \sin O \cos T \cos S] + \\ & Z_p(\cos O \sin R - \sin O \sin T \cos R) \\ Y_c = & X_p[(-\sin O \cos R + \cos O \sin T \sin R)\cos S - \cos O \cos T \sin S] + \\ & Y_p[(-\sin O \cos R + \cos O \sin T \sin R)\sin S + \cos O \cos T \cos S] + \\ & Z_p(-\sin O \sin R - \cos O \sin T \cos R) \\ Z_c = & X_p(-\cos T \sin R \cos S - \sin T \sin S) + \\ & Y_p(-\cos T \sin R \sin S + \sin T \cos S) + \\ & Z_p(\cos R \cos T) \end{aligned}$$

A.3 Counterclockwise from East (CCE) Aiming

When Type C photometry with CCE aiming, the following equations will convert between photometric angles and a photometric coordinate system:

$$\begin{aligned} H &= \text{atan2}(Y_p, X_p) \text{ (See Section A.7, Notes 2, 3, 5.)} \\ V &= \text{atan2}[\sqrt{X_p^2 + Y_p^2}, -Z_p] \text{ (See Section A.7, Note 3.)} \\ X_p &= \cos H \sin V \\ Y_p &= \sin H \sin V \\ Z_p &= -\cos V \end{aligned}$$

When using Type B photometry with CCE aiming, the following equations will convert between photometric angles and a photometric coordinate system:

$$\begin{aligned} H &= \text{atan2}[-Y_p, \sqrt{X_p^2 + Z_p^2}] \text{ (See Section A.7, Notes 2, 3.)} \\ V &= \text{atan2}(X_p, -Z_p) \text{ (See Section A.7, Notes 3, 4.)} \\ X_p &= \cos H \sin V \\ Y_p &= -\sin H \\ Z_p &= -\cos H \cos V \end{aligned}$$

Conversion from space coordinates to photometric coordinates with CCE aiming can be made using the following equations:

$$\begin{aligned} X_p = & X_c(\cos O \cos T \cos S + (-\sin O \cos R - \cos O \sin T \sin R)\sin S) + \\ & Y_c(\sin O \cos T \cos S + (\cos O \cos R - \sin O \sin T \sin R)\sin S) + \\ & Z_c(\sin T \cos S + \cos T \sin R \sin S) \\ Y_p = & X_c(-\cos O \cos T \sin S + (-\sin O \cos R - \cos O \sin T \sin R)\cos S) + \\ & Y_c(-\sin O \cos T \sin S + (\cos O \cos R - \sin O \sin T \sin R)\cos S) + \\ & Z_c(-\sin T \sin S + \cos T \sin R \cos S) \\ Z_p = & X_c(\sin O \sin R - \cos O \sin T \cos R) + \end{aligned}$$

$$Y_c(-\cos O \sin R - \sin O \sin T \cos R) + \\ Z_c(\cos T \cos R)$$

Conversions from photometric coordinates to space coordinates with CCE aiming can be made with the following equations:

$$\begin{aligned} X_c &= X_p(\cos O \cos T \cos S + (-\sin O \cos R - \cos O \sin T \sin R)\sin S) + \\ &\quad Y_p(-\cos O \cos T \sin S + (-\sin O \cos R - \cos O \sin T \sin R)\cos S) + \\ &\quad Z_p(\sin O \sin R - \cos O \sin T \cos R) \\ Y_c &= X_p(\sin O \cos T \cos S + (\cos O \cos R - \sin O \sin T \sin R)\sin S) + \\ &\quad Y_p(-\sin O \cos T \sin S + (\cos O \cos R - \sin O \sin T \sin R)\cos S) + \\ &\quad Z_p(-\cos O \sin R - \sin O \sin T \cos R) \\ Z_c &= X_p(\sin T \cos S + \cos T \sin R \sin S) + \\ &\quad Y_p(-\sin T \sin S + \cos T \sin R \cos S) + \\ &\quad Z_p(\cos R \cos T) \end{aligned}$$

A.4 Convert between CCE and CN Aiming Systems

Converting between aiming systems is easy. Developers may choose to have programs operate internally with just one coordinate system but use the user's preferred system for interaction, input, and output. (See **Section A.7**, Note 1.)

$$\begin{aligned} O_{CCE} &= 90 - O_{CN} & O_{cn} &= 90 - O_{CCN} \\ T_{CCE} &= T_{CN} & T_{cn} &= T_{CCE} \\ R_{CCE} &= R_{CN} & R_{cn} &= R_{CCE} \\ S_{CCE} &= -S_{CN} & S_{cn} &= -S_{CCE} \end{aligned}$$

A.5 Converting from “Aiming at a Point” to an Angular (CN or CCE) System

Under the “aiming a luminaire at a point” positioning system, x_a , y_a , and z_a are the “global” coordinates of the “aiming point” in the space, with the origin at the luminaire (photometric) center.

The Orientation angle (O) and Tilt angle (T) for equivalence with the CN positioning system can be found from the following equations:

$$\begin{aligned} O &= \text{atan2}(x_a, y_a) \text{ (See Section A.7, Note 3.)} \\ T &= \text{atan2}[\text{sqrt}(x_a^2 + y_a^2), -z_a] \text{ (See Section A.7, Note 3.)} \end{aligned}$$

The Orientation angle (O) and Tilt angle (T) for equivalence with the CCE positioning system can be found from the following equations:

$$\begin{aligned} O &= \text{atan2}(y_a, x_a) \text{ (See Section A.7, Note 3.)} \\ T &= \text{atan2}[\text{sqrt}(x_a^2 + y_a^2), -z_a] \text{ (See Section A.7, Note 3.)} \end{aligned}$$

A.6 Converting Four-Angle Aiming into Three-Angle Aiming

Formulae can be developed to convert four-angle aiming (Orientation, Tilt, Roll, and Spin) into one of the three-angle forms (Orient, Tilt, and Spin or Orient, Tilt, and Roll). While ongoing efforts have been reported from computer programmers currently developing such software, there is as yet no complete, published monograph available for example or reference.

A.7 Notes for Annex A

1. Angles are listed here in degrees, since the photometric standard also lists angles in degrees. Specific implementations on specific libraries may require that radians or other angle measures be used.
2. If X_p , Y_p , and Z_p are all 0, no corresponding photometric angles can be determined.
3. The photometric expression $\text{atan2}(A, B)$ means: the arc tangent of A/B . It is required to yield a four-quadrant result (either 0° to 360° or -180° to 180°) and should choose proper quadrants when A or B is 0. If A and B are both 0, a result of 0° can be used. (See Note 4.)
4. In Type B photometry, any vertical angle can be used when the horizontal angle is -90° or 90° . Candela values at horizontal angles of 90° or -90° should (in theory) be identical for all vertical angles.
5. In Type C photometry, any horizontal angle can be used when the vertical angle is 0° or 180° . Candela values at vertical angles of 0° and 180° should (in theory) be identical for all horizontal angles.

Annex B – Overview of Goniometer Types A, B, and C

The Type A and Type B goniometers are described in *ANSI/IES LS-4-20, Lighting Science: Measurement of Light – The Science of Photometry*.⁸ They arise from different types of equipment and are fundamentally different in the photometric web that they produce. Type A produces a web having the poles at nadir and zenith, while Type B produces the poles in a horizontal plane.

The Type C goniometer is different from Type A and Type B mechanically, but produces the photometric web as Type A.

Type C is the form in common use in the United States (although it was formerly referred to as Type A).

As described in ANSI/IES LS-4-20,⁸ Type A would normally use a vertical angle of 0 degrees for the aiming direction of the floodlight (a horizontal line). This is a line aimed at the equator and not at the pole. On Type C goniometers, however, it is conventional to designate nadir as 0° vertical. Hence, the two systems differ by 90° .

It should be noted that it is the IES recommended procedure that nadir be referred to as 0° . If this is the case, then it does matter if the goniometer is Type A or Type C.

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Order# ANSI/IES LM-72-20

ISBN# 978-0-87995-215-0

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